

Hoan Nguyen

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EDUCATION

Kenyon College

Expected Graduation: May 2028

Bachelor of Arts in Mathematics and Statistics

GPA: 4.0/4.0

- **Awards:** Merit List (all semesters), Presidential Scholarship, 4th Prize in Vietnam Science and Engineering Fair
- **Coursework:** Bayesian Analysis*, Data Analysis, Software Development*, Data Structures and Program Design, Elements of Statistics, Linear Algebra*, Multivariable Calculus, Calculus II, Introduction to Programming

WORK EXPERIENCE

Vietcombank

Jun. 2025 – Aug. 2025

Data Science Intern - Risk and Quant Department

Hanoi, Vietnam

- Engineered 9 new time-window and ratio features in **SQL** for credit risk modeling, which included implementing data quality checks and **reducing the daily pipeline runtime by 18%** on datasets with **250K+ rows**
- **Executed** a periodic Probability of Default (PD) model refresh by **training** a regularized logistic regression in Python (pandas, scikit-learn, and 5-fold CV), improving AUC from **0.78 to 0.82**
- **Collaborated** with a fellow intern to design **2** interactive Power BI dashboards, giving teams a work performance overview and rapid geographic credit risk insights

Hanoi University of Science and Technology

Sep. 2024 – Dec. 2024

Data Science Research - Image-Based PM2.5 Prediction Project

Remote

- Developed an **image preprocessing pipeline** for 1,200+ sky images using Python (PIL, NumPy) to optimize GPU memory for **deep learning model training**
- **Engineered** 3 geo-temporal features, **fusing** them with image records and partitioning the dataset (70/15/15 split) for **rigorous model validation**
- **Combined** 3 engineered image features (entropy, depth, residual) with 3 geo-temporal fields to create **6-dimensional input** vectors for multimodal learning
- Developed and **optimized** a hybrid regression model (3-layer MLP + LR) in PyTorch, achieving an **85% explained variance** and significantly **outperforming** the SVR baseline

Kenyon College

Oct. 2024 – May 2025

Technical Support Assistant - LBIS

Gambier, OH

- **Performed** data cleaning and validation for **500+ inventory records** in Excel, using pivot tables and formulas to identify discrepancies and achieve **98%** data integrity
- **Designed and built 2** interactive **Power BI dashboards** to visualize printer usage and device lifecycle data, providing **actionable insights** to support IT procurement strategies for 4 campus departments

PROJECTS

Financial Complaint Outcome Predictor | DuckDB, XGBoost, Streamlit, Scikit-learn

[Website](#)

- **Executed** robust preprocessing pipeline for 300K+ consumer complaints, using **Target Encoding** for high-cardinality features and ADASYN for class imbalance, **expanding the dataset by 118%**
- **Engineered** multi-class classifier (benchmarking XGBoost, CatBoost, Random Forest) to predict corporate responses for the top 9 US financial institutions, reaching **72% weighted precision**
- **Deployed** full-stack ML application (Streamlit/Python) with a modular backend, enabling **real-time prediction** capabilities for **non-technical** stakeholders

End-to-end Car Price Predictor | Scikit-learn, XGBoost, Pandas, NumPy, Streamlit

[Website](#)

- Performed large-scale EDA and **data cleaning** (480k+ → 410k+ rows) and built an automated scikit-learn **preprocessing pipeline** with ColumnTransformer for 13 mixed-type features
- Trained, **tuned** (using RandomizedSearchCV), and evaluated multiple machine learning regressors (Ridge, Random Forest, XGBoost), selecting the optimized **XGBoost** model for its superior **86% R^2** score

CFPB Analytics Dashboard | Streamlit, Plotly, Pandas, dbt, DuckDB

[GitHub](#)

- **Engineered** an **ELT pipeline** (dlt, DuckDB, dbt, Prefect) for **10** big companies with incremental loads, **9** dbt models to extract CFPB complaints into analytics-ready models
- **Developed** an interactive Streamlit **dashboard** analyzing **340K+** complaints to visualize key performance metrics
- Identified **2** critical product **friction points** by performing root cause analysis via Plotly **treemaps**

TECHNICAL SKILLS

Languages: Python, R, SQL

Libraries: Pandas, NumPy, SciPy, Scikit-learn, Keras, Statsmodels, Scikit-learn, PyTorch, TensorFlow, XGBoost, CatBoost, Matplotlib, Seaborn, ggplot2, Shiny

Databases/Tools: AWS, Azure, DuckDB, Jupyter Notebook, Excel, Power BI, Tableau, MySQL, RStudio